

Adversarial Robustification via Text-to-Image Diffusion Models

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01 Robustness Gap and Limitations

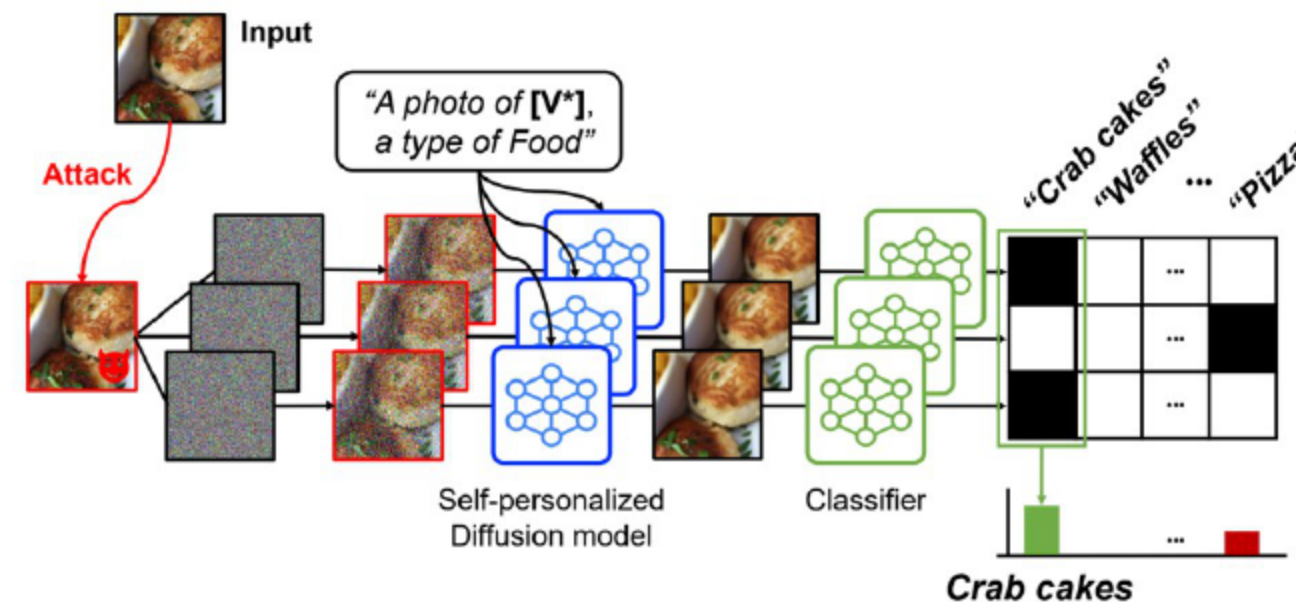
- Scaling helps natural accuracy, yet **adversarial robustness lags**: off-the-shelf models remain attack-prone
- Prevailing defenses need **specialized training** on large labeled data, incurring **clean versus robust trade-offs**
- Training data is often **unavailable** for public pretrained models, blocking downstream robustification
- Prior transfer approaches (adversarial CLIP fine-tuning) require ImageNet-scale data and **overfit to domains**
- Goal: **robustify pretrained classifiers without training data**, preserve clean accuracy, and provide **certified robustness**

[Coates et al. 2011; Wu et al. 2020]

02 Data-Free Robustification via Denoised Smoothing

- Use **text-to-image super-resolution diffusion** as a task-agnostic denoiser within denoised smoothing
- Zero-shot text conditioning with class prompts; **no external images required**
- **k-timestep correction**: align noisy inputs with super-resolution dynamics for higher fidelity
- Synthesize a few **class-referenced images** to enable self-personalization of the denoiser
- Fine-tune the classifier on denoised synthetic samples to close the clean versus denoised gap

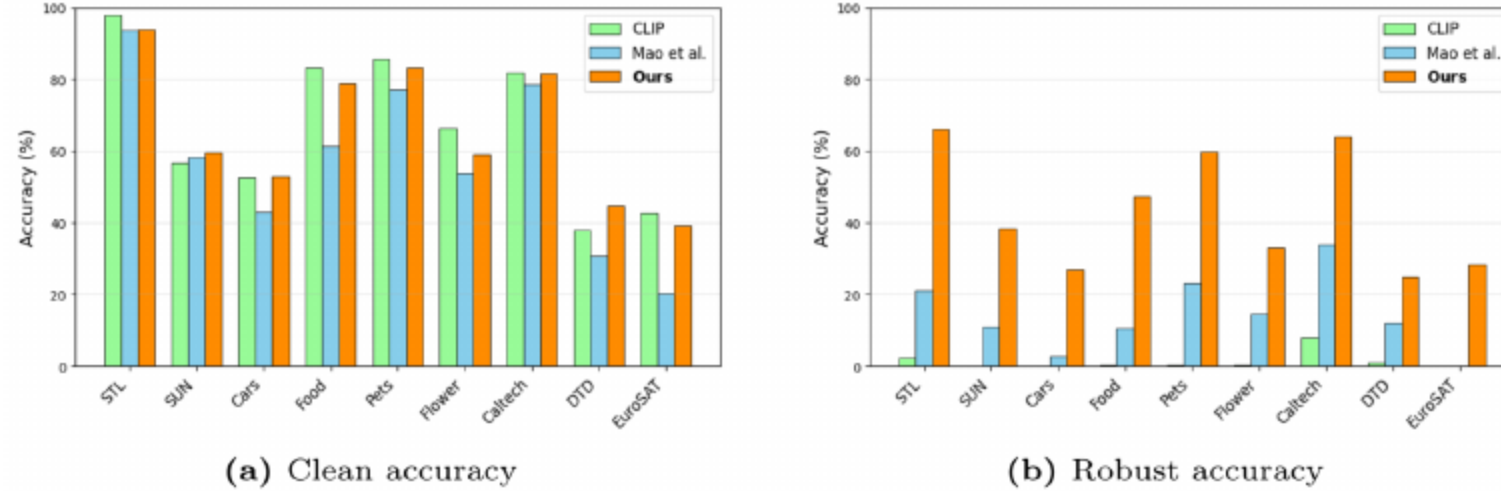
03 Single Visual Anchor



- Denoise-and-classify via randomized smoothing with **text-conditioned super-resolution denoiser**
- **Self-adaptation** using text-only references: personalize denoiser and lightly tune classifier
- **Provable certificates** from smoothing; scalable to many off-the-shelf classifiers

04 Teaser: Frontier Gains

2 D. Choi et al.



- **New robustness versus accuracy frontier** for CLIP in zero-shot settings
- **Certified robust accuracy** surpasses data-intensive baselines across datasets
- Generalizes to non vision-language models, for example ResNet-50, with **competitive or better certified robustness**

05 Randomized Smoothing Essentials

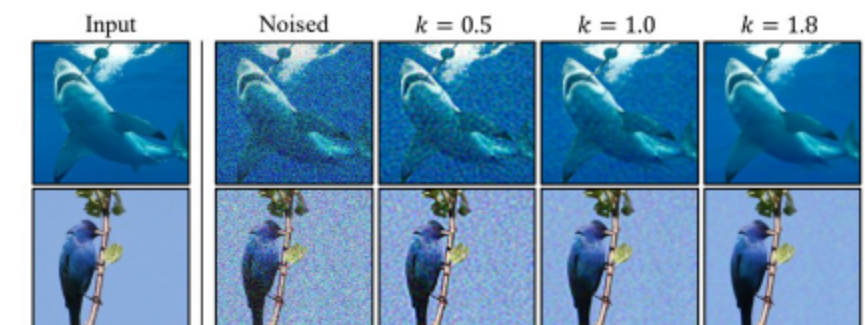
- Smoothed classifier: majority vote over predictions on Gaussian noised inputs
- Certified radius grows with the probability gap between top and runner-up classes
- Denoised smoothing: **prepend a denoiser** to lift accuracy under noise before classification

06 Why Super-Resolution Diffusion

- **Pixel-space cascaded text-to-image** models directly denoise inputs, unlike latent-only designs
- Super-resolution stages are biased toward **reconstruction** over free-form generation, preserving semantics
- Text conditioning enables **zero-shot, task-aware denoising** from class names

07 Timestep Mapping and k-Correction

- Map input noise level to a diffusion timestep and self-condition with a scaled noisy input
- Use a **higher auxiliary timestep** with factor k greater than one to attenuate excess high frequencies
- Super-resolution models expect upsampled low-frequency priors; **k-correction** restores denoising fidelity and certificates



- Denoiser: **text-conditioned super-resolution diffusion** with a class prompt template
- Zero-shot uses an empty prompt; after personalization, use a unique token for task adaptation
- Certification via standard randomized smoothing estimation on the composed model

- Generate a small **text-only reference set**: a few images per class using the class prompt
- DreamBooth-style **personalization** on the super-resolution denoiser with classifier-guided regularization
- Fine-tune the classifier on denoised synthetic images to reduce distribution shift

	Method	STL	SUN	Cars	Food	Pets	Flower	DTD	Caltech	Average
-	CLIP	97.8	56.8	52.7	83.0	85.7	66.3	37.8	81.9	70.3
$\varepsilon = 0.5$	CLIP-Smooth	75.0	46.8	42.1	52.3	66.7	43.5	17.2	68.3	51.5
	Mao et al. [37]	94.8	60.0	48.7	69.7	80.8	57.7	34.0	79.7	65.7
	Ours (Certified)	94.8 (90.4)	58.6 (55.4)	54.1 (49.7)	80.2 (74.5)	83.6 (81.9)	61.4 (58.7)	42.7 (38.9)	81.7 (79.1)	69.6 (66.1)
$\varepsilon = 1.0$	CLIP-Smooth	32.4	27.7	40.8	31.3	43.8	36.8	7.0	54.0	34.2
	Mao et al. [37]	93.4	58.2	42.9	61.2	77.0	53.6	30.8	78.5	62.0
	Ours (Certified)	93.8 (80.2)	59.4 (53.6)	52.9 (45.5)	78.8 (64.8)	83.1 (77.7)	58.9 (48.3)	39.1 (32.4)	81.7 (75.7)	68.5 (59.8)

- **Higher robust accuracy** than CLIP-Smooth and adversarially fine-tuned CLIP
- **Certified accuracy** at target budgets exceeds empirical robustness of baselines on most datasets
- Maintains or improves clean accuracy; offers **flexible trade-offs** across robustness budgets

		Robust accuracy (%)			Clean accuracy (%)		
	Method	CropDisease	EuroSAT	ISIC	CropDisease	EuroSAT	ISIC
$\varepsilon = 0.5$	CLIP	0.0	0.0	0.0	20.9	<u>42.6</u>	<u>27.3</u>
	CLIP-Smooth	1.8	<u>13.6</u>	<u>9.0</u>	7.1	16.6	14.3
	Mao et al. [37]	<u>2.2</u>	0.6	4.2	16.0	25.0	26.9
	Ours (Certified)	11.5 (4.3)	29.0 (11.0)	17.6 (5.8)	<u>20.3</u> (16.8)	45.2 (39.0)	35.7 (33.7)
$\varepsilon = 1.0$	CLIP	0.0	0.0	0.0	20.9	<u>42.6</u>	27.3
	CLIP-Smooth	<u>1.2</u>	<u>11.8</u>	<u>4.6</u>	4.3	18.2	10.6
	Mao et al. [37]	0.6	0.0	2.2	<u>16.0</u>	20.4	22.0
	Ours (Certified)	4.9 (0.8)	28.2 (5.0)	8.4 (1.4)	15.8 (8.4)	44.8 (37.8)	<u>26.3</u> (22.0)

- Consistent robustness on specialized datasets; prior transfer methods **fail under shift**
- Competitive on ImageNet **without any training data** compared to data-dependent methods
- **Self-adaptation** contributes notable certified gains on ImageNet

Method	Data-free?	Robust accuracy (%)		Clean accuracy (%)	
		$\epsilon = 0.5$	$\epsilon = 1.0$	$\epsilon = 0.5$	$\epsilon = 1.0$
Standard Training	✗	5.2	1.0	74.4	74.4
+ Ours (w/o adapt)	✓	<u>56.2</u> (47.0)	44.2 (27.4)	<u>73.0</u> (67.0)	68.8 (57.2)
+ Ours	✓	57.0 (50.4)	47.8 (34.0)	70.4 (68.2)	<u>71.8</u> (60.8)
Adversarial Training [36]	✗	51.0	<u>46.8</u>	55.0	55.0
Randomized Smoothing [13]	✗	55.2 (48.6)	43.8 (37.0)	65.4 (66.8)	55.4 (57.0)
Carlini et al. [7]	✗	<u>56.2</u> (49.2)	45.2 (33.2)	72.6 (67.4)	70.0 (57.8)

- Data-free robustification **surpasses adversarial training** and prior denoised smoothing under certification
- Large robust gains with **minimal clean accuracy drop** versus standard training
- Adaptation further **boosts certified accuracy** at higher robustness budgets

σ	k	ACR	Certified accuracy at ϵ (%)					
			0.0	0.25	0.5	0.75	1.0	1.25
0.25	0.5	0.152	36.2	23.0	14.3	6.6		
	1.0	0.229	46.0	35.6	25.2	13.4		
	1.8	0.277	50.8	41.8	29.6	19.2		
0.50	0.5	0.080	13.2	10.2	6.0	4.3	3.4	2.0
	1.0	0.200	28.8	22.0	16.4	11.2	8.4	6.0
	1.8	0.367	42.0	35.4	29.2	23.8	17.6	13.0

- Increasing k yields **clearer denoised outputs** and larger certified radii
- Certified accuracy improves markedly across noise scales and budgets with k near one point eight
- k compensates the **frequency mismatch** inherent to super-resolution conditioning

ϵ	Adapt.		Certified accuracy at ϵ (%)			
	T2I	CLIP	ImageNet	STL	SUN	Food
0.5	✗	✗	29.6	55.2	28.3	43.6
	✓	✗	<u>31.8</u>	66.0	<u>30.7</u>	<u>43.8</u>
	✓	✓	34.2	66.0	32.1	45.7
1.0	✗	✗	17.6	27.0	17.3	21.8
	✓	✗	<u>19.4</u>	<u>40.8</u>	<u>19.3</u>	<u>27.3</u>
	✓	✓	20.6	41.2	22.5	28.9

- Personalizing the super-resolution denoiser **raises certified accuracy** consistently
- Classifier fine-tuning on denoised synthetic samples adds **further gains**
- A single adapted pipeline benefits **multiple noise levels** via randomized training timesteps

- Empirical defenses via adversarial training versus **certified defenses** under smoothing
- Zero-shot recognition with vision and language models and robustness transfer
- Data-free robustification enabled by **text-conditioned diffusion denoisers**
- Contrasts with **data-heavy transfer** and dataset-specific diffusion training
- Scales to **diverse classifiers** beyond CLIP

- **Data-free, certified robustification** via text-conditioned super-resolution diffusion denoisers
- k-timestep correction and **self-adaptation** are crucial components
- Generalizes beyond CLIP with **strong empirical and certified gains** across domains
- Practical path to robustify off-the-shelf models **without access to training data**
- Code: <https://github.com/ChoiDae1/robustify-T2I>



THANKS!